- Goal:

<u>Thm</u> Every set of the form $[a_1,b_1]\times\ldots\times[a_n,b_n]$ is compact in $\mathbb{R}^n$ (with the metric $d_p$ for any $1\le p\le\infty$).

- Last time: step 1

<u>Prop</u> If $[a_k,b_k]\subseteq\mathbb{R}$ are intervals s.t.

$$[a_k,b_k]\supseteq[a_{k+1},b_{k+1}]\quad\forall k\in\mathbb{N},$$

then $\bigcap_{k\in\mathbb{N}}[a_k,b_k]\neq\emptyset$.

<u>Cor</u> Let $n\in\mathbb{N}$. If the sets

$$C_k=[a_{k,1},b_{k,1}]\times\ldots\times[a_{k,n},b_{k,n}]\subseteq\mathbb{R}^n$$

satisfy $C_k\supseteq C_{k+1}\ \forall k\in\mathbb{N}$, then $\bigcap_{k\in\mathbb{N}}C_k\neq\emptyset$.

<u>Pf</u>: Apply the Proposition to each component $[a_{k,m},b_{k,m}]$.
Get: $\exists\ x_m\in\bigcap_{k\in\mathbb{N}}[a_{k,m},b_{k,m}]$ for each $m=1,2,\ldots,n$.
Then $x=(x_1,\ldots,x_n)\in\bigcap_{k\in\mathbb{N}}C_k$. □

- Now we're ready to prove:

<u>Thm</u> Every set of the form $[a_1,b_1]\times\ldots\times[a_n,b_n]$ is compact in $\mathbb{R}^n$.

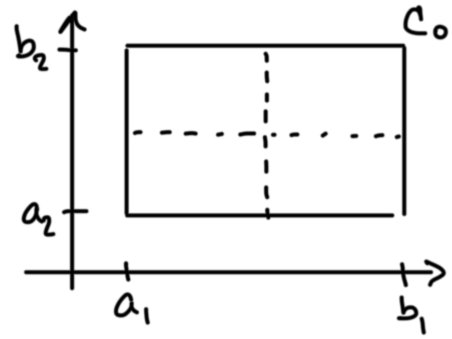
<u>Pf</u>: Suppose not: there is some $C_0=[a_1,b_1]\times\ldots\times[a_n,b_n]$ that is not compact. Then $\exists$ an open cover $\{U_i\}_{i\in I}$ with no finite subcover.

Bisect each of the intervals:

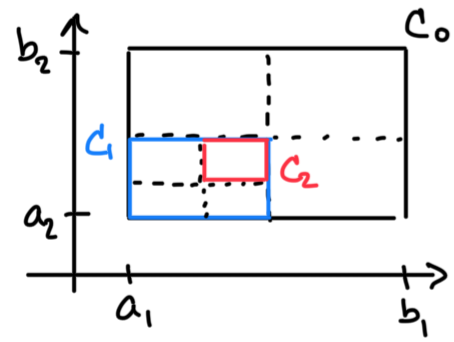
$$[a_1,b_1]=\left[a_1,\tfrac{a_1+b_1}{2}\right]\cup\left[\tfrac{a_1+b_1}{2},b_1\right]$$

⋮

$$[a_n, b_n] = \left[a_n, \frac{a_n+b_n}{2}\right] \cup \left[\frac{a_n+b_n}{2}, b_n\right]$$




This subdivides $C_0$ into $2^n$ sets. At least one of these sets, call it $C_1$, cannot be covered by finitely many $\{U_i\}_{i\in I}$. (Otherwise, all $2^n$ sets could be covered by finitely many $\{U_i\}_{i\in I}$, and then $C_0$ would have a finite subcover.)




Similarly, we bisect each side of $C_1$. One of these $2^n$ subsets, call it $C_2$, cannot be covered by finitely many $\{U_i\}_{i\in I}$.

Proceeding inductively, we get sets

$$C_k = [a_{k,1}, b_{k,1}] \times \dots \times [a_{k,n}, b_{k,n}] \subseteq \mathbb{R}^n$$

s.t.:

- $C_k \supseteq C_{k+1} \quad \forall k$
- $C_k$ cannot be covered by finitely many $\{U_i\}_{i\in I}$
- The sides of $C_k$ have length

$$b_{k,1} - a_{k,1} = \frac{b_1 - a_1}{2^k}, \quad \dots, \quad b_{k,n} - a_{k,n} = \frac{b_n - a_n}{2^k}.$$

By the Corollary, $\exists\, x \in \bigcap_{k\in\mathbb{N}} C_k \subseteq C_0$. As $\{U_i\}_{i\in I}$ is a cover of $C_0$, $\exists\, i_0 \in I$ s.t. $x \in U_{i_0}$.

As $\mathcal{U}_{i_0}$ is open, $\exists r > 0$ s.t. $B_r(x) \subseteq \mathcal{U}_{i_0}$. Choose $k \geq 1$ large enough s.t.

$$\text{largest side of } C_k = \frac{\max_{m=1,\dots,n} (b_m - a_m)}{2^k} < r.$$

Then $x \in C_k \subseteq B_r^{d_\infty}(x) \subseteq \mathcal{U}_{i_0}$.

$y \in C_k \Rightarrow d_\infty(x,y) = \max_{j=1,\dots,n} |x_j - y_j| < r$

$\Rightarrow y \in B_r^{d_\infty}(x)$

But then $\{\mathcal{U}_{i_0}\}$ is a finite subcover for $C_k$, which contradicts how we chose $C_k$. $\square$

## Connected Sets

<u>Def</u> Let $(X,d)$ be a metric space and $A, B \subseteq X$. We say that $A$ and $B$ are <u>separated</u> if:

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

<u>Rmk</u> Separated sets are disjoint: $A \cap B \subseteq \overline{A} \cap B = \emptyset$.
But disjoint sets need not be separated: e.g., $A = (-1, 0)$ and $B = [0, 1)$ in $(\mathbb{R}, |\cdot|)$ are...

- disjoint: $A \cap B = \emptyset$
- not separated: $\overline{A} \cap B = \{0\} \neq \emptyset$

<u>Def</u> Let $(X,d)$ be a metric space and $A \subseteq X$. We say:

① $A$ is <u>disconnected</u> if $\exists$ nonempty separated sets $B, C \subseteq X$ s.t. $A = B \cup C$.

② A is <u>connected</u> if it is not disconnected.

<u>Ex</u> $A = \underbrace{(-1,0)}_{=B} \cup \underbrace{(0,1)}_{=C}$ is disconnected in $(\mathbb{R}, |\cdot|)$:

Set

- $A = B \cup C$
- B and C are nonempty
- B and C are separated:

$$\overline{B} \cap C = [-1,0] \cap (0,1) = \emptyset$$
$$B \cap \overline{C} = (-1,0) \cap [0,1] = \emptyset$$

<u>Prop</u> Let $(X, d_X)$ be a metric space and $(Y, d_Y)$ a subspace. Then: Y is connected in $(X, d_X)$ $\Longleftrightarrow$ Y is connected in $(Y, d_Y)$.

<u>Pf</u>: $\Rightarrow$|: Suppose not: Y is connected in X but Y is disconnected in Y. Then $\exists A, B \subseteq Y$ s.t.:

- $Y = A \cup B$
- $A \neq \emptyset$, $B \neq \emptyset$
- $\overline{A}^Y \cap B = \emptyset$, $A \cap \overline{B}^Y = \emptyset$

Recall that $\overline{A}^Y = \overline{A}^X \cap Y$ by lecture 16. So:

$$\emptyset = \overline{A}^Y \cap B = \overline{A}^X \cap \underbrace{Y \cap B}_{=B} = \overline{A}^X \cap B$$

Similarly,

$$\emptyset = A \cap \overline{B}^Y = A \cap Y \cap \overline{B}^X = A \cap \overline{B}^X$$

Therefore A and B are also separated in X, and so $Y$ is disconnected in $X$ — a contradiction.

$\underline{\Leftarrow}$: Suppose not: $Y$ is connected in $Y$, but $Y$ is disconnected in $X$. Then $\exists C, D \subseteq X$ s.t.:

- $Y = C \cup D$
- $C \neq \emptyset$, $D \neq \emptyset$
- $\overline{C}^X \cap D = \emptyset$, $C \cap \overline{D}^X = \emptyset$

Then

$$\overline{C}^Y \cap D = \overline{C}^X \cap Y \cap D = \underbrace{\overline{C}^X \cap D}_{=\emptyset} \cap Y = \emptyset$$

$$C \cap \overline{D}^Y = \underbrace{C \cap \overline{D}^X}_{=\emptyset} \cap Y = \emptyset$$

Therefore $C$ and $D$ are separated in $Y$, and so $Y$ is disconnected in $Y$ — a contradiction. $\square$